

Improper integrals – 4

Principal value, symmetry and regularisation

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Theoretical minimum.

Definition (principal value). For a singularity at c , v. p. $\int_a^b f = \lim_{\varepsilon \rightarrow 0^+} (\int_a^{c-\varepsilon} f + \int_{c+\varepsilon}^b f)$, and at infinity v. p. $\int_{-\infty}^{\infty} f = \lim_{R \rightarrow \infty} \int_{-R}^R f$. The two sides are made to cancel each other on purpose — which is exactly what the definition of a convergent improper integral forbids. Thus v. p. $\int_{-1}^1 \frac{dx}{x} = 0$ while the integral itself diverges.

Theorem 1 (when the principal value exists). If f is C^1 in a neighbourhood of c , then v. p. $\int_a^b \frac{f(x)}{x-c} dx$ exists, since $\frac{f(x)-f(c)}{x-c}$ is bounded and $\frac{f(c)}{x-c}$ is odd about c . Symmetry is the only mechanism here: the principal value exists precisely when the divergent parts on the two sides are equal and opposite.

Theorem 2 (regularisation by subtraction). If $\int_0^{\infty} f$ and $\int_0^{\infty} g$ both diverge but $f - g$ is integrable, the difference $\int_0^{\infty} (f - g)$ is a legitimate object and is the standard way to attach a value to a divergent expression. Frullani's integral $\int_0^{\infty} \frac{f(ax)-f(bx)}{x} dx = (f(0) - f(\infty)) \ln \frac{b}{a}$ is exactly of this shape, and so is $\int_0^{\infty} (\frac{1}{e^x-1} - \frac{1}{x}) e^{-x} dx$.

Fact 1 (the Borwein threshold). If $a_1 \geq a_2 + \dots + a_n$ with all $a_k > 0$, then

$$\int_{-\infty}^{\infty} \prod_{k=1}^n \frac{\sin a_k x}{a_k x} dx = \frac{\pi}{a_1},$$

and as soon as the inequality is violated the value drops. The reason is that the Fourier transform of the product is a convolution of boxes, which stays flat at the origin exactly until the widths add up past a_1 .

Fact 2 (what symmetry can and cannot do). Oddness gives the principal value for free but never gives convergence; a change of variable that maps the domain onto itself may prove that two divergent halves cancel, and nothing

more. Before using symmetry one has to know that the integral converges — otherwise one proves, as is classical, that $0 = \ln 2$.

1. Prove Theorem 1, compute v. p. $\int_{-1}^1 \frac{dx}{x}$, v. p. $\int_{-\infty}^{\infty} \frac{x dx}{1+x^2}$ and v. p. $\int_0^2 \frac{dx}{x-1}$, and show that none of the three integrals converges. Then give an example of an odd function for which even the principal value does not exist.
2. Prove that $\int_0^{\infty} \frac{e^{-x} - \cos x}{x} dx = 0$ and $\int_0^{\infty} \left(\frac{1}{1+x} - e^{-x} \right) \frac{dx}{x} = \gamma$, and explain both as instances of Theorem 2.
3. (МФТИ) Let $a_k = 2^{1-k}$. Find the value of the integral $\int_{-\infty}^{+\infty} \prod_{k=1}^n \frac{\sin a_k x}{a_k x} dx$ as a function of $n \geq 2$.
4. Prove Fact 1 for $n = 2$ and $n = 3$ directly, and then show that $\int_0^{\infty} \prod_{k=0}^n \frac{\sin \frac{x}{2^{k+1}}}{\frac{x}{2^{k+1}}} dx = \frac{\pi}{2}$ for $n \leq 6$ and is strictly smaller for $n = 7$.
5. (Капачёв 1.75) Does there exist a non-zero infinitely differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that for every polynomial p the integral $\int_{-\infty}^{+\infty} p(x)f(x) dx$ converges and equals zero?